



DCA 4 Guide and Installation

Papercut integration and Device Web access



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Introduction

The **DCA 4** is the latest version of the award-winning **DCA of MPS Monitor** that from the design point of view represents a breaking point being the result of a new and revolutionary project.

It's important to remember that the DCA (Data Collection Agent) of MPS Monitor has over **half a million installations** and is currently used by thousands of dealers to manage **1.2 million devices** in over **60 countries**. Success due to innovative features such as the ability to configure DCAs in clusters, integration with SDS services for monitoring and management of HP devices and ease of installation and configuration.

So why are we launching a new DCA? The answer is a mix between improving the performance of the previous version (e.g. optimizing data flow, improving troubleshooting and security) and introducing, thanks to the use of the latest IOT protocols, a series of revolutionary features that are not currently found on the market:

- **Real-time connection** between user, agent and device thanks to MQTT protocol, IOT protocol for TCP/IP data transmission
- **Access to the device web page (Device Web Access)** without the need for a remote connection to the customer's network (see chapter *Device Web Access*)
- **High security** via SNMP 3 network protocol and credential management
- **Granular data collection** that allows you to set the frequency of updates for each type of data (metrics, consumable levels, alerts, etc.)
- **Papercut integration** without any license and additional costs (see chapter *Papercut integration*)
- **New device discovery process** that provides greater efficiency and stability while also simplifying the troubleshooting process

In the next chapters we will see in detail how to download, install and configure the DCA 4 and what are the necessary requirements, plus we will focus on the most important news of the new agent, from integration with Papercut to Device web access.

Hardware requirements (Windows)

The recommended hardware configuration for the DCA connector for Windows is shown below:

RAM: 2 GB

CPU: Intel Core 3

HD space: at least 300 MB

The PC does not need to be dedicated, and can be used simultaneously for other uses. The **DCA 4 connector** works totally in the background and PC users are not affected in any way by the use of other applications.

The PC must be equipped with Internet access via HTTPS and be able to reach the printers to be managed, via TCP and UDP protocols.



For installations in **small and medium-sized** networks from one to a few dozen printers), configuration with recommended **hardware requirements** is ample enough to scan in seconds or at most in minutes (depending on how many devices are present).

For large networks, if you have a lot of printers to scan (hundreds or thousands) and require short scanning times, it is valid the rule **best hardware -> better performance**

The DCA connector has a scanning speed proportional to the available hardware and the extent to which it is intended to be used. It is possible, on very large networks and with a high-power hardware, to build a configuration with high parallelism, which drastically reduces the time of data detection.

Hardware requirements (Linux)

The recommended hardware configuration for the DCA connector for Linux is shown below:

RAM: 2 GB

CPU: Intel Core 3

HD space: at least 300 MB

The PC does not need to be dedicated and can be used simultaneously for other uses. The **DCA 4 connector** works totally in the background and PC users are not affected in any way by the use of other applications.

The PC must be equipped with Internet access via HTTPS and be able to reach the printers to be managed, via TCP and UDP protocols.



For installations in **small and medium-sized** networks (from one to a few dozen printers), configuration with **recommended hardware** requirements is ample enough to scan in seconds or at most in minutes (depending on how many devices are present).

For **large networks**, if you have a lot of printers to scan (hundreds or thousands) and require short scanning times, it is valid the rule **best hardware -> better performance**

The DCA connector has a scanning speed proportional to the available hardware and the extent to which it is intended to be used. It is possible, on very large networks and with high-power hardware, to build a configuration with high parallelism, drastically reducing the data detection time.

For the Raspberry, the minimal hardware required is Raspberry PI 3.

Hardware requirements (macOS)

The minimum hardware configuration for macOS is the hardware supported by macOS 12 Monterey or higher. At least 600 MB free on the storage.

The Mac does not need to be dedicated and can be used simultaneously for other uses. The **DCA 4 connector** works totally in the background and PC users are not affected in any way by the use of other applications.

The PC must be equipped with Internet access via HTTPS and be able to reach the printers to be managed, via TCP and UDP protocols.



For installations in **small and medium-sized** networks (from one to a few dozen printers), configuration with **recommended hardware** requirements is ample enough to scan in seconds or at most in minutes (depending on how many devices are present).

For **large networks**, if you have a lot of printers to scan (hundreds or thousands) and require short scanning times, it is valid the rule **best hardware -> better performance**

The DCA connector has a scanning speed proportional to the available hardware and the extent to which it is intended to be used. It is possible, on very large networks and with a high-power hardware, to build a configuration with high parallelism, drastically reducing the data detection time.

Software requirements (Windows)

The DCA connector shall operate on a system equipped with one of the following operating systems:

Windows 10 64-bit build 1607 (E) or higher

Windows Server 2016 64-bit or higher

For details, refer to the operating systems supported by Microsoft .NET on which the DCA is based.

<https://github.com/dotnet/core/blob/main/release-notes/8.0/supported-os.md#windows>

The DCA does not require any other software component (e.g. Database, Web server, etc.), as all data management and web application functions are delegated to the portal.

Software requirements (Linux)

The DCA connector shall operate on a system equipped with one of the following operating systems:

Debian 12 or higher, Ubuntu 20.04 or higher, Raspberry Pi OS (Arm32/Arm64)

For details, refer to the operating systems supported by Microsoft .NET on which the DCA is based.

<https://github.com/dotnet/core/blob/main/release-notes/8.0/supported-os.md#linux>

The DCA does not require the installation or configuration of any other software components (e.g. Database, Web server, etc.), as all data management and web application functions are delegated to the portal.

Software requirements (macOS)

The DCA connector shall operate on a system equipped with one of the following operating systems:

macOS 12 (Monterey) or higher

For details, refer to the operating systems supported by Microsoft .NET on which the DCA is based..

<https://github.com/dotnet/core/blob/main/release-notes/8.0/supported-os.md#apple>

The DCA does not require the installation or configuration of any other software components (e.g. Database, Web server, etc.), as all data management and web application functions are delegated to the portal.

Safety requirements

The DCA connector must be hosted on a system with stable access to the following Web endpoints:

Web endpoints	Protocols and ports
https://*.mpsmonitor.com	HTTPS, port 443
*.mpsmonitor.com	SSH, port 22
https://*.mpsmonitor.com	HTTP2 GRPC, port 443
*.mpsmonitor.com	MQTT over WSS, port 443 <i>(deprecated)</i>
*.mpsmonitor.com	MQTT, port 8883 <i>(deprecated)</i>
eu01-broker-mpsmonitor.azure-devices.net	MQTT, port 8883
eu01-broker-mpsmonitor.azure-devices.net	MQTT over WSS, port 443

HP JAMC Web endpoints requirements	Protocols and ports
https://jamanagement.hp.com	HTTPS, port 443
https://connectivity.pod1.avatar.ext.hp.com	HTTPS, port 443
https://ews.hpjamservices.com	HTTPS, port 443

HP JAMC Web endpoints requirements	Protocols and ports
http://crl.sca1b.amazontrust.com/sca1b.crl	HTTP, port 80
https://registration.pod1.avatar.ext.hp.com	HTTPS, port 443
http://crl3.digicert.com/DigiCertGlobalG2TLSRSA2562020CA1.crl	HTTP, port 80
http://crl4.digicert.com/DigiCertGlobalG2TLSRSA2562020CA1.crl	HTTP, port 80
https://eu.jamanagement.hp.com	HTTPS, port 443
https://eu.ews.hpjamservices.com	HTTPS, port 443
http://signal.pod1.avatar.ext.hp.com	HTTP, port 80
http://crl3.digicert.com/DigiCertGlobalG2TLSRSA2562020CA1-2.crl	HTTP, port 80
http://crl4.digicert.com/DigiCertGlobalG2TLSRSA2562020CA1-2.crl	HTTP, port 80



Please note that depending on the country where the DCA is installed, access to some of the listed domains may not be necessary.

If access to the Internet via HTTPS is managed through a proxy server:

- If the proxy does not require specific authentication, the connector does not need any configuration, as it is able to recognize and interface with the proxy automatically.
- If the proxy requires authentication credentials, you must have a proxy user that is enabled to reach the above domains.
- The proxy must support the HTTP2 GRPC protocol.

Any specific Proxy user will be inserted during the configuration of the connector via the local interface of the DCA.

The credentials (user and password) for HTTPS access via proxy are saved by the connector, in encrypted form, only and exclusively on the customer's system; are never transmitted to the portal or to other sites outside the customer's network, and are never visible in clear to the staff who access the customer's system. In the customer's internal network, the following local ports must allow communication between the PC or server where the DCA connector is installed and the IP addresses of the printers:

Port	Protocol	Port	Protocol
161	UDP	3910	TCP
427	UDP	3911	TCP
443	TCP	7627	TCP
3702	UDP	8080	TCP

The connector installation procedure installs on the host system two Windows system services, which are automatically configured and by default use the Local System user for the "Monitor" service (which performs all DCA maintenance tasks) and Network Service for the "Client" service (which scans the printer's local network and all the necessary monitoring operations).

If the network and the host system have no special restrictions on the Local System user, no change to the default configuration is required, nor is any additional requirement required.

If, on the other hand, there are particular restrictions or policies that prevent the activation of the Local System user or the execution of the service, a specific local user will be required.

Ensure that your client's firewall and security infrastructure allow you to run the application and reach the domains needed by your country.

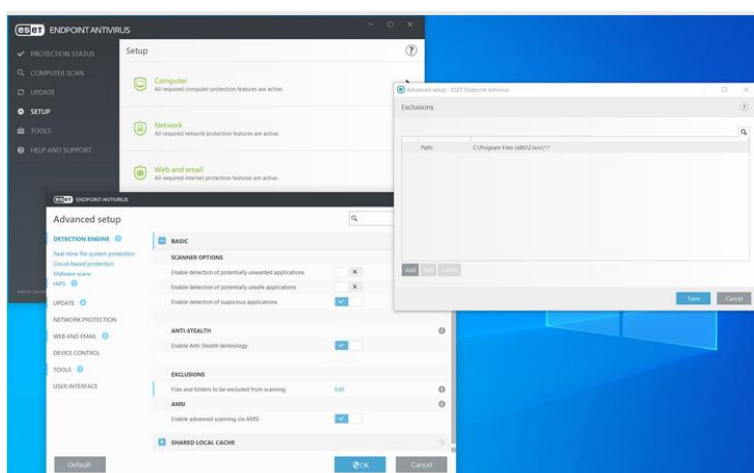
If necessary, contact our support team for help identifying the right security configuration to enable DCA operations for a specific customer.

Antivirus e Firewall

Antivirus is software that can block the execution of some programs installed on the PC or server in order to limit the entry into the system of malware or malicious codes that can damage the system.

Some antivirus software will likely recognize the DCA connector as a threat and block it. To allow the DCA connector to communicate the data collected by the printers and to save them in the cloud, you must include in the antivirus exclusion list the location where the DCA connector was installed, in this case, you should exclude from the antivirus block the following paths C: Program Files MpsMonitor DcaConnector and C: Program Files (x86) HP JetAdvantage Management. This is the standard path on which the software installs itself during automatic installation. To avoid blocking the DCA connector you should also exclude these files: client/MpsMonitorDca.Client.exe, monitor/MpsMonitorDca.Monitor.exe, client/MpsMonitorDca.Identity.exe.

Below is an exclusion screen from antivirus blocking taken from one of the most common software in this category:



The Firewall is a perimeter defense component of a computer network that performs functions of connection between two or more segments of the network, providing, therefore, a protection in terms of cybersecurity of the network itself. This service is already present within operating systems such as Windows and allows within a company, or a home network, for example, to block outgoing or incoming traffic on some ports or services.

Installation of the DCA 4

The DCA connector is located between the various components of the system. The DCA connector is installed on the PC or server in the customer's LAN and performs the reading operations following the instructions provided by the server component on the portal. The information is transmitted to server components using the HTTPS protocol.

Important



The custom installation package should only be used for the customer to whom it refers, to allow the system to properly save the data in the archive. In case of multiple installations or if you need to reinstall the package, it must be downloaded from the site at the time of use and should not be saved and used later.

If you create a customer, you can download a custom installation package for that specific customer. During the installation process, you will always see the customer code/description: always make sure to install the corresponding package for the specific customer.

To install the DCA connector on the customer's computer, direct access to the customer's computer will be required either via virtual machine or with the help of remote access software.

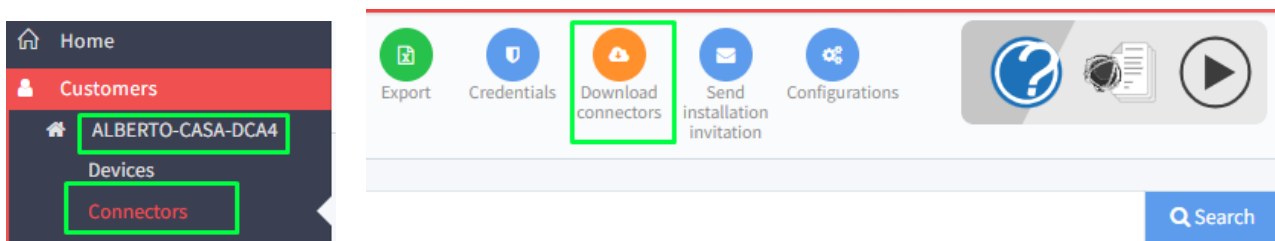


On Windows operating systems you can install the DCA v4 on the same machine where the previous Windows connector is already installed (explorer 3)

Download of the DCA 4

In this chapter we will follow the path to download the DCA 4. The first step is to select the client on which you want to install the connector and then click on **Connectors** in the submenu of the selected customer (image 1). Now select "Download connectors" in the top right menu.

Image 1



In the new page select the operating system for which you want to do download the connector. You can choose between **Windows, Linux, Raspberry, macOS**. If you choose Windows, be sure to have the DCA 4 submenu selected.

You can click on the blue button or, alternatively, you can copy the link from the portal and send it by email (image 2). A new page will open, informing us that we are downloading the DCA for the chosen customer and the selected operating system (image 3-5).

Image 2

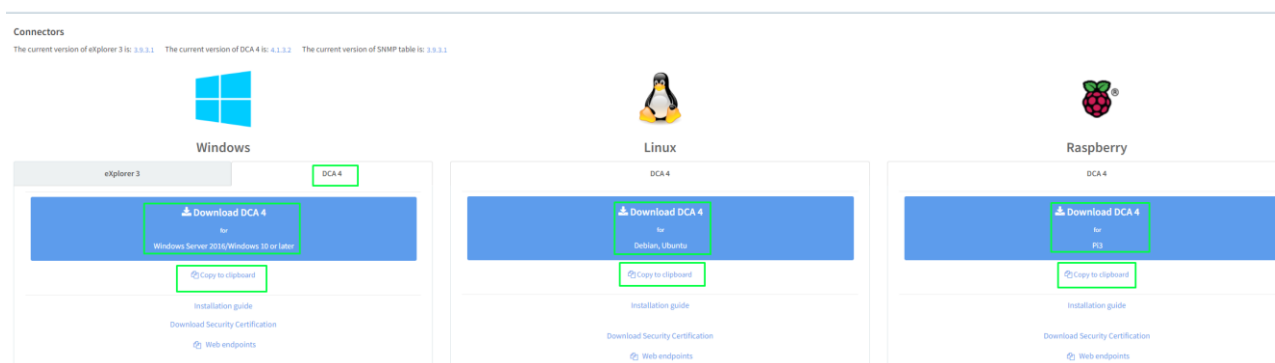


Image 3

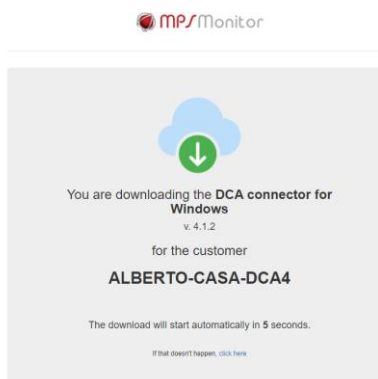


Image 4

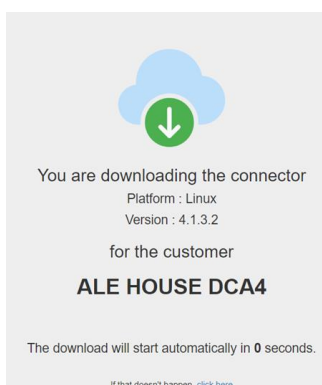
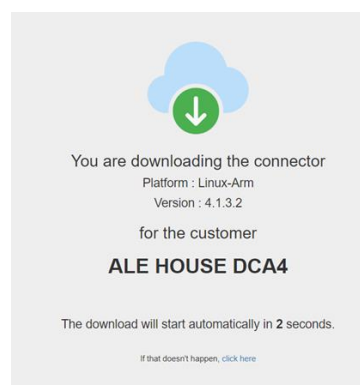


image 5



1. Setup generation error

If the download procedure fails to generate the setup package, the following message will appear.



In the footer there is a link that allows you to generate a support request by e-mail to help@mpsmonitor.com. The request is already pre-filled with the object, which allows you to immediately identify the problem.

To	help@mpsmonitor.com
Cc	
Subject	Problema installazione DCA Windows - Cliente: XXXXXXXX - Dealer MPS Monitor srl - Token 71ce64c9-42a5-4e54-8391-feceb510310b

2. Invalid link error

If the download link is invalid (for example, it was not copied correctly), the following message will appear.



In the footer there is a link that allows you to generate a support request by e-mail to help@mpsmonitor.com. The request is already pre-filled with the object, which allows you to immediately identify the problem.

To	help@mpsmonitor.com
Cc	
Subject	Problema installazione DCA Windows - Token 71ce64c9-42a5-4e54-8391-feceb510323

Installation of the DCA 4 for Windows

To install the DCA 4 connector, do the following:

- Download the DCA connector configuration from the link provided after creating the customer location of the DCA connector on the MPS Monitor portal. The link can only be generated by a user with administrator or installer credentials on the MPS Monitor portal. For more information on how to create the link, see the Configuration Guide and the Technical Reference for the DCA connector, or contact your MPS Monitor system provider to request it.
Or download the DCA connector configuration by following the Download DCA for Windows steps above.

- Double click on setup exe will appear this message: Allow the following program to make changes to this computer? Click on Yes (Yes)



Always check the correctness of the Customer's identification number and description. If you use a package that does not match the Customer on which you want to install it, cancel the installation and get the link to download the installation package successfully

- Accept the **terms of the license agreement** and click Next (Image 2)

Image 1

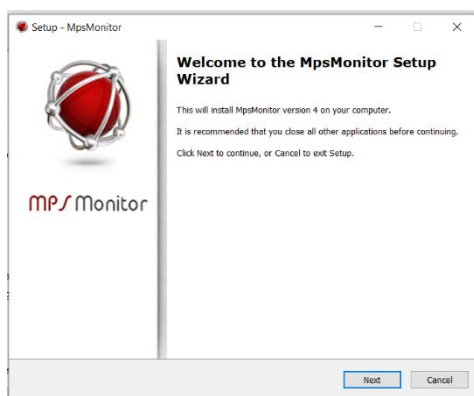
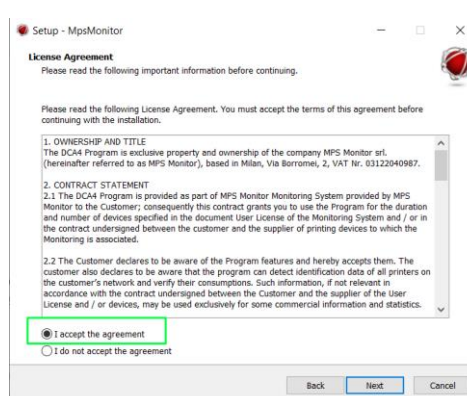


Image 2



- Click on next to download **HP JetAdvantage** Management Connector (image 3)

- Only in case the agent JAMC has been previously installed - Click on **continue anyway** (image 4)

Image 3

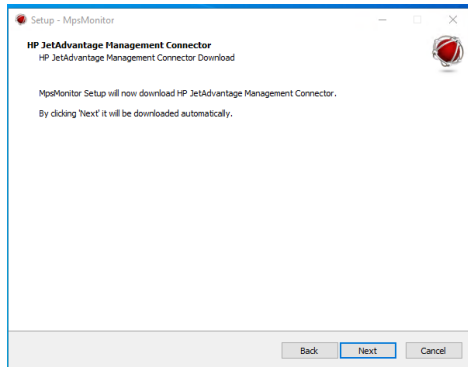
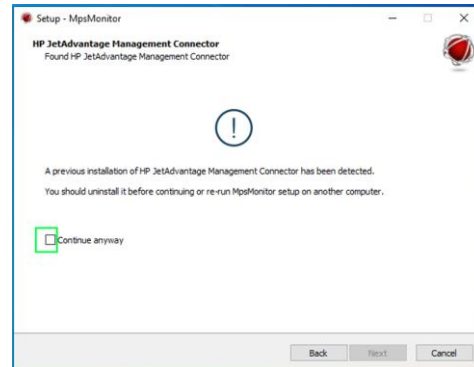


Image 4



- By accepting the agreement (image 5) the setup will start the download (image 8)

Image 5

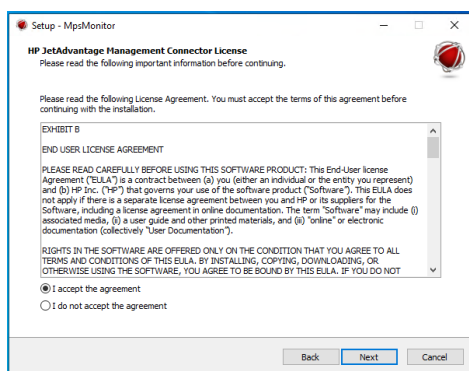
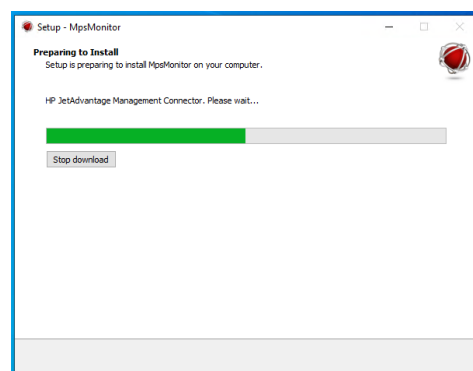


Image 6



- Select the **additional Tasks** that will run during the installation and click Next (image 9)
- At this point the **program is ready for installation**, click on Install (image 10)

Image 7

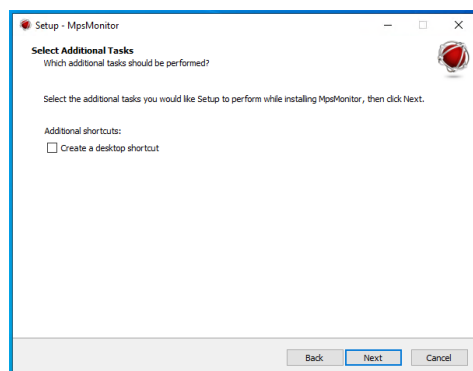
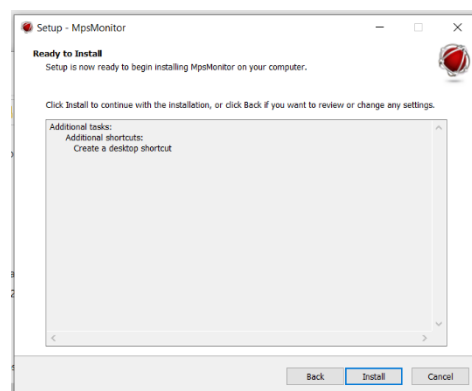


Image 8



- When the **installation is complete**, click Finish (image 10)

Image 9

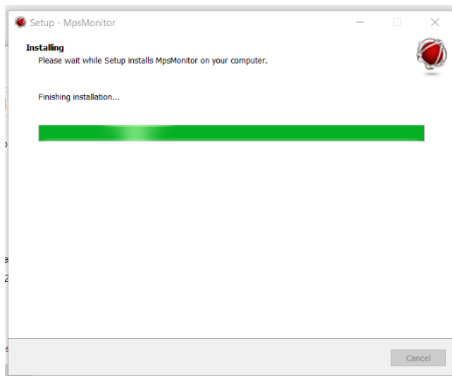
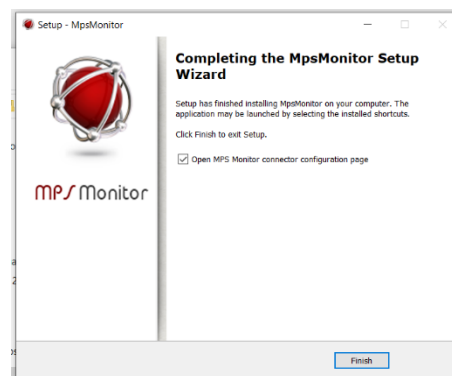


Image 10



The upgrade from a version of the DCA connector to a higher version is performed automatically, without the need to access the customer's system, as soon as a new version is published. If you want to turn off automatic updates on the DCA connector, please contact Technical Support at help@mpsmonitor.com.

At the end of the installation, if you have flagged "Open *MPS Monitor connector configuration page*" (image 11), you will access the **connector configuration page** where you can navigate between **Home** (image 12), **Managed devices** (image 13), and **Configurations** (image 14).

Image 11

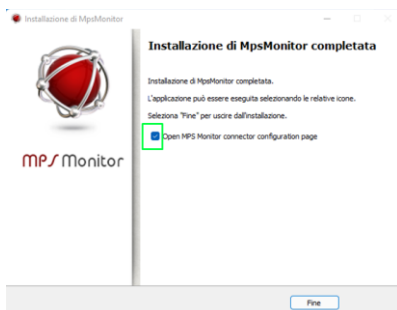


Image 12

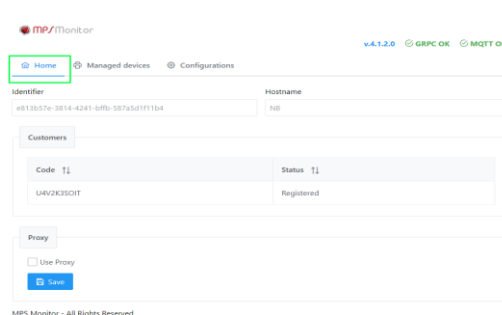
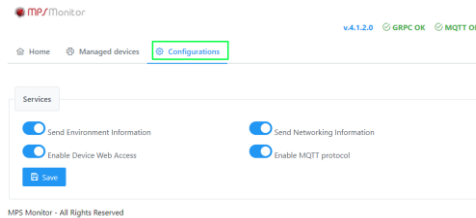
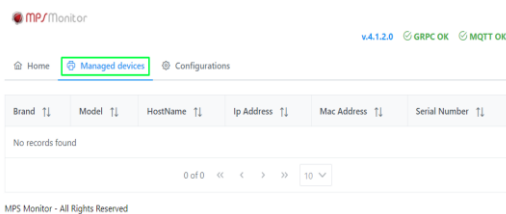


Image 13

Image 14



You can still open the **connector configuration** page, after installation, at any time by connecting with any browser to the following address: <http://localhost:12312/home> and from this page you can see the following information:

- Customers configured
- the connector status (GRPC server in the upper right)
- printers detected

In the next chapter, we will see how to configure a possible network proxy server.

Installation of the DCA 4 for Linux

In this guide, we download the setup in the /tmp folder.

```
sudo chmod +x MpsMonitor.Dca.Setup_<DATE>.<CUSTOMERCODE>.<STACK>.run  
sudo ./MpsMonitor.Dca.Setup_<DATE>.<CUSTOMERCODE>.<STACK>.run
```

```
root@ubuntu2204:/tmp#
root@ubuntu2204:/tmp# chmod +x MpsMonitor.Dca.Setup_20241016115814.SWLCAWTJ46.eu.pkg
root@ubuntu2204:/tmp# ./MpsMonitor.Dca.Setup_20241016115814.SWLCAWTJ46.eu.pkg
Creating directory MpsMonitor.Dca.Setup
Verifying archive integrity... 100% MD5 checksums are OK. All good.
Uncompressing MPS Monitor DCA4 setup 100%

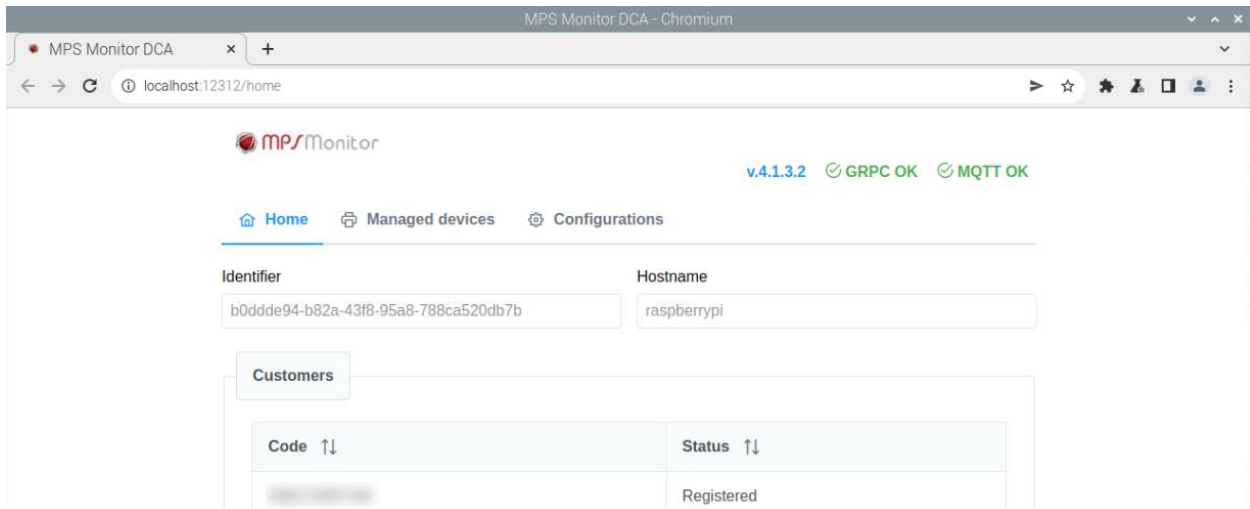
.....
Stating MPS Monitor DCA 4 installation
The installation will create a new user mpsmonitoruser
and will install the software in the user home directory,
the MPS Monitor service will be started with this user as well
.....

./install.sh: 21: python: not found
Do you wish to install MPS Monitor DCA 4 for the customer with code [SWLCAWTJ46]? (type Y or N)|
```

```
Done!
Copy installation package
Done!
Convert Windows Line Endings to Unix
Done!
Add mpsmonitoruser in the sudoers group and add the permission to start and stop MPS Monitor DCA services
mpsmonitoruser ALL= NOPASSWD: /bin/systemctl start mpsmonitordcaclient
mpsmonitoruser ALL= NOPASSWD: /bin/systemctl stop mpsmonitordcaclient
mpsmonitoruser ALL= NOPASSWD: /bin/systemctl restart mpsmonitordcaclient
mpsmonitoruser ALL= NOPASSWD: /bin/systemctl start mpsmonitordcamonitor
mpsmonitoruser ALL= NOPASSWD: /bin/systemctl stop mpsmonitordcamonitor
mpsmonitoruser ALL= NOPASSWD: /bin/systemctl restart mpsmonitordcamonitor
Done!
Setup appsettings
EU endpoint
Replace CustomerCode, Identifier and Url
Done!
Change the ownership of the files to mpsmonitoruser
Done!
Set the permissions to executables
Done!
Link the services definitions to etc systemd system
Done!
Reload service definitions and enable the services
Done!
Start MPS Monitor service
Done!
.....
Installation completed!
Access the page http://localhost:12312/home to check your configuration.
.....
root@ubuntu2204:/tmp# |
```

The install script should end with “Installation completed!”

Goto to this web page to check the DCA registration: <http://localhost:12312/home>



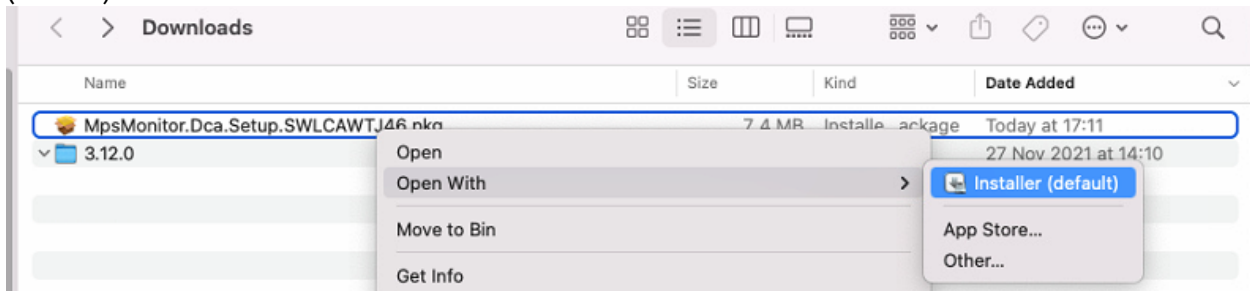
Installation of the DCA 4 for macOS

WARNING: the setup name contains the CUSTOMER CODE necessary for registration, do not rename it and check that there are no spaces or other characters due to system renaming.

The software is not downloaded from the Apple Store so the operating system will display this warning:



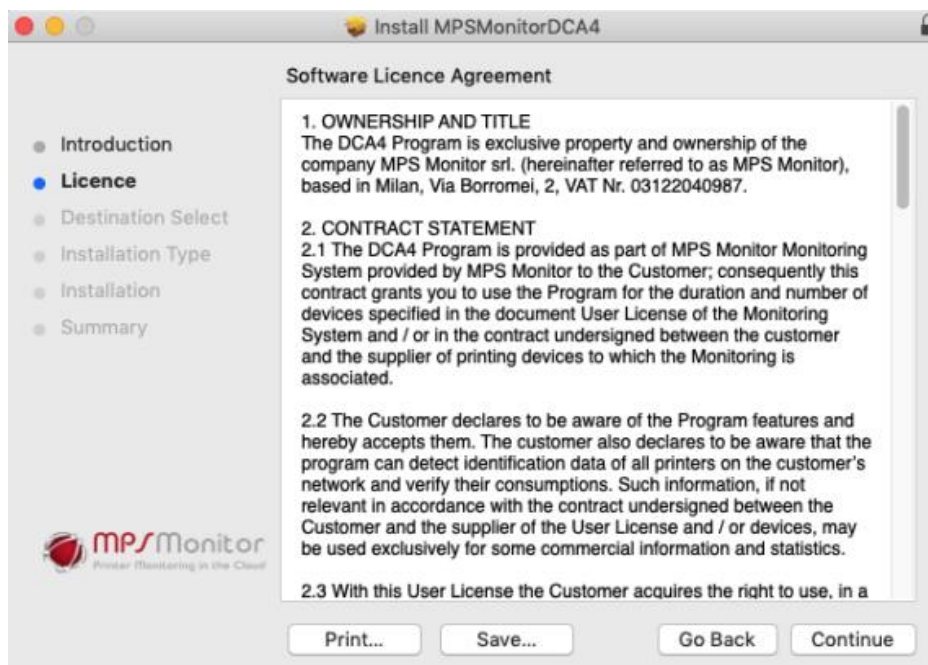
In order to continue with the installation, you need to right-click (or hold down the Control button and click with the mouse) the installation file, select "Open with" and then select "Installer (default)":



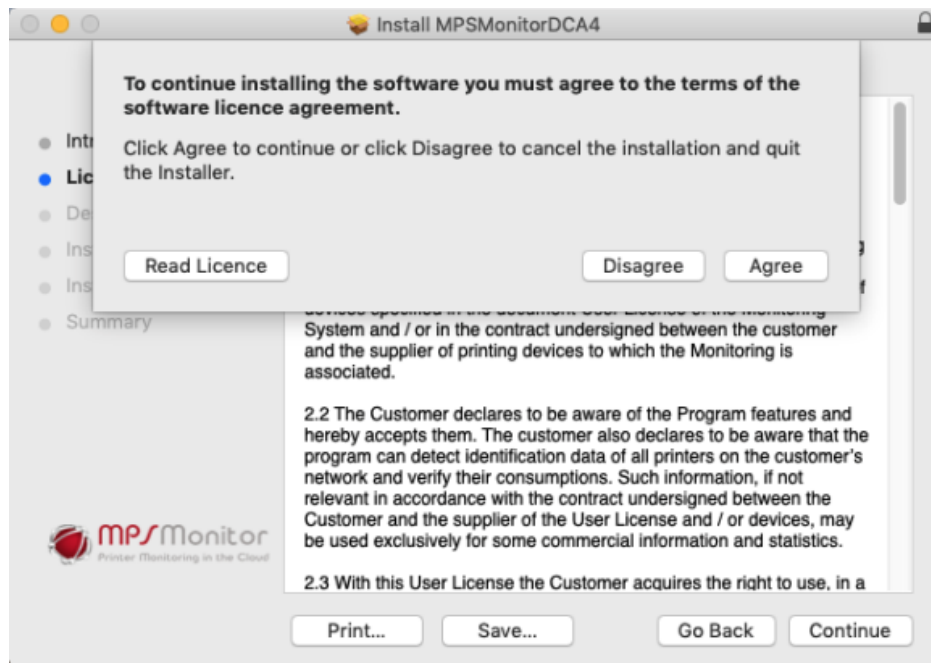
This will display the previous warning but will give you the option to continue by clicking "Open":



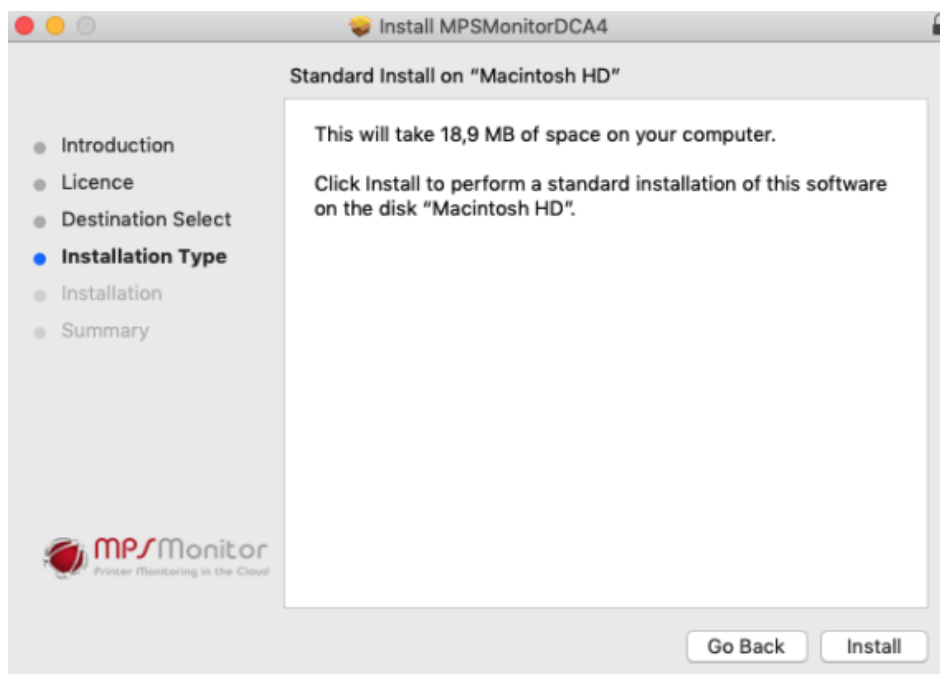
At this point the installation proceeds as follows:



License needs to be accepted to continue with the installation:



Click "Install" to install the software and finish the installation.



http://localhost:12312/home

MPS Monitor - All Rights Reserved

Proxy setting

Proxy setting using the configuration utility

Configuring the proxy using the program <DIR INSTALLAZIONE>/client/ MpsMonitorDca is possible. ConfigurationUtility.

On the various operating systems, it will be necessary to have administrative permissions and in particular:

- On Linux operating systems, you need to run the command as the mpsmonitoruser user, then run the following command:

```
sudo runuser -l mpsmonitoruser -c '/home/mpsmonitoruser/MpsMonitor/DcaConnector/client/MpsMonitorDca.ConfigurationUtility'
```

- On OSX operating systems, you must run the command as the mpsmonitoruser user, then run the following command:

```
sudo -H -u mpsmonitoruser bash -c '/Users/mpsmonitoruser/MpsMonitor/DcaConnector/client/MpsMonitorDca.ConfigurationUtility'
```

The console will show the following options, press 1 to start the setup..

```
-----
MpsMonitorDca Configuration Utility
-----

Choose an option from the following list:

1 - Setup Proxy Configuration
2 - Remove Proxy Configuration
3 - Configure Connector Features
q - Exit
```

Proxy setup using web interface (after first registration)

If the connector has already been registered on the portal, you can configure it from the **configuration page of the connector** that we access by connecting to <http://localhost:12312/home> (see previous chapter) and putting a checkmark next to *Use Proxy* (image 1), you access the **Proxy configuration page** (image 2).

Image 1

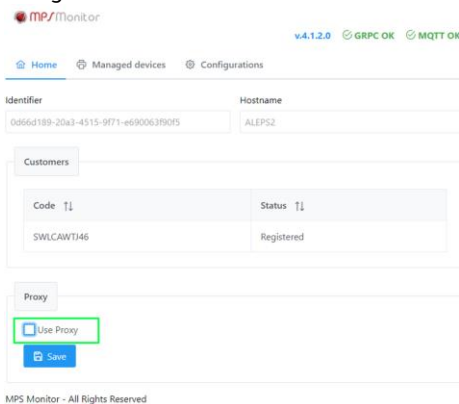
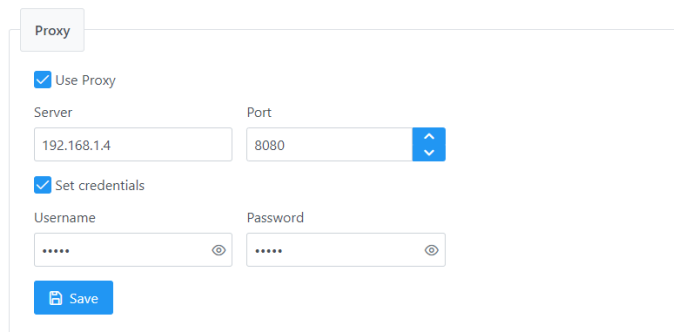


Image 2



At this point you will be asked for an **OTP code** to confirm the operation (image 3), which can be recovered by moving to the **Connectors** section of the customer's page and clicking on **View OTP Connector** (image 4).

Image 3

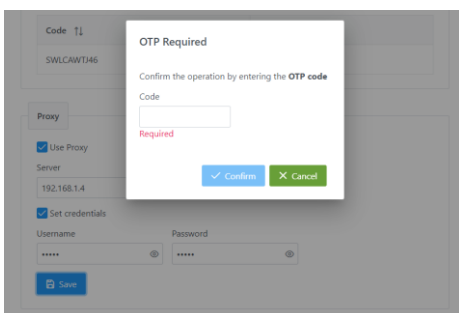
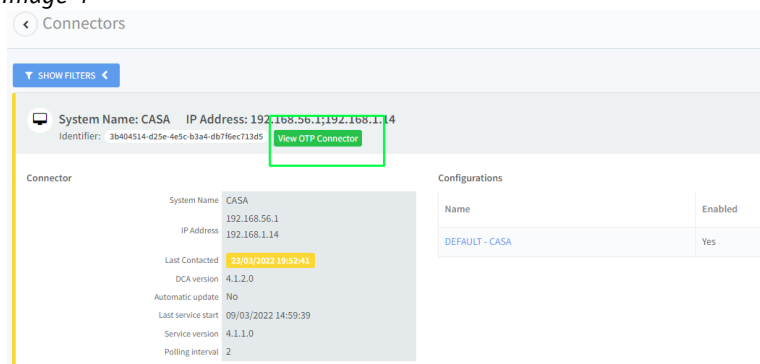


Image 4



DCA 4 configuration

The configuration of the connector on the portal has been revised bringing countless advantages:

- **Inserted the scan intervals (in minutes) instead of the schedules** - in the **Scan intervals** tab it is possible to set different scanning time intervals for specific types of data such as meter reading, consumable reading, etc. (image 1). In the **Working days** tab, users can set windows in which data collection can take place. You can set different windows for different days. By default, the system returns an 8am - 21 pm interval for working days only (image 2).

Image 1

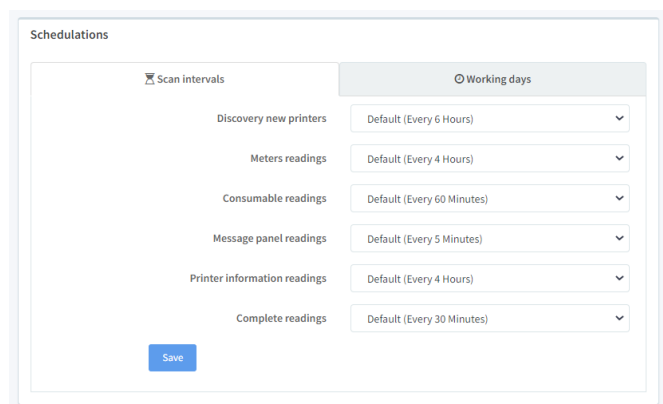
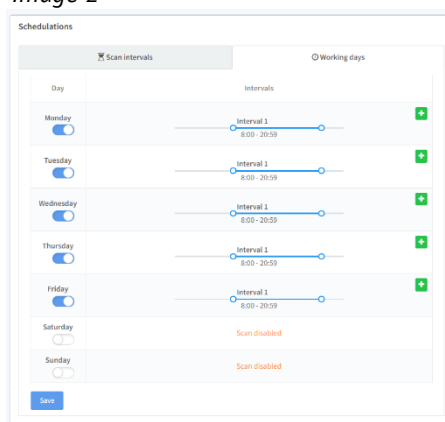


Image 2



- From this version you can **enter the hostnames** to monitor. In large networks where the DHCP protocol is used, you do not need to scan the entire network. Just add the list of hostnames. The hostnames configuration can be added together with the IP Address configuration (image 3). In fact, through the "Add" button you can insert one or more hostnames, one per line (image 4).

Image 3

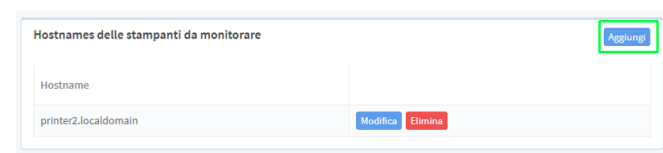


Image 4



- **Removed the community field** - from this new version, the credentials are set on a special page that we will see in the next understood

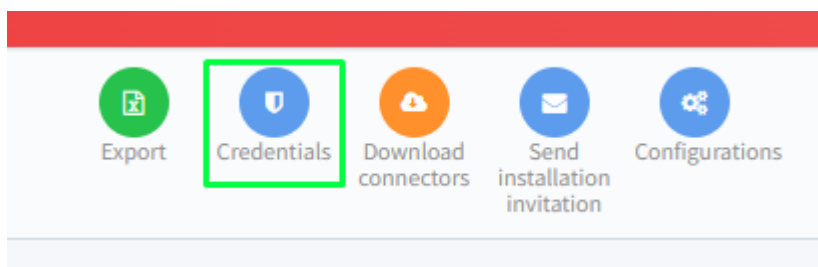
DCA 4 credential management

In the connectors page (Home > Customer > Connectors) has been added the **Credentials** button that allows you to access the management page of the customer's **credential set** (image 1).

The set of credentials of a client is the set of passwords used in the client's network (be it community for **SNMPv1v2** protocol or **SNMPv3** credentials).

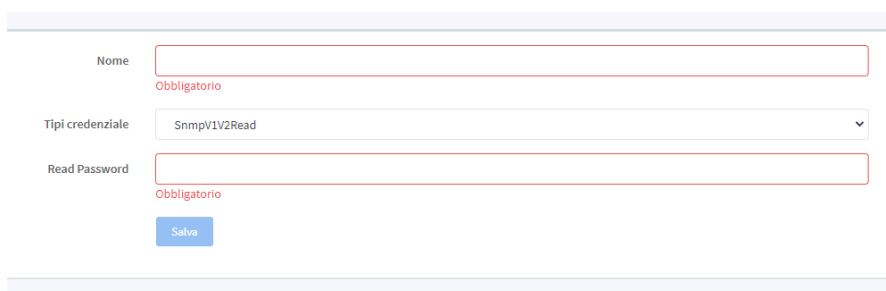
These sets will then be used sequentially while scanning the customer's network for the correct printer credential.

Image 1



To add a new credential select **New credentials**, enter a name for the set and select the type of credential (image 2).

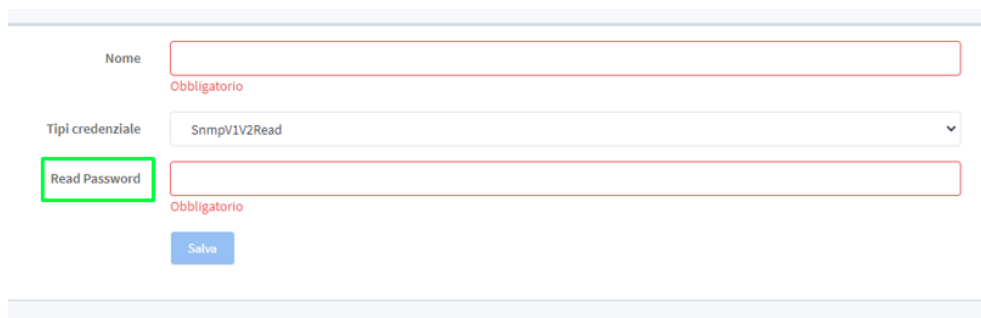
Image 2

A screenshot of a web form for adding new credentials. It has a light blue header bar. The form contains three input fields: 'Nome' (Name) with a red border and the word 'Obbligatorio' (Mandatory) in red below it; 'Tipi credenziale' (Credential type) with a dropdown menu showing 'SnmpV1V2Read'; and 'Read Password' with a red border and 'Obbligatorio' in red below it. At the bottom of the form is a blue button labeled 'Salva' (Save).

SNMPV1V2 Read

In the Read Password field enter the **community** (image 3)

Image 3



Nome

Obbligatorio

Tipi credenziale

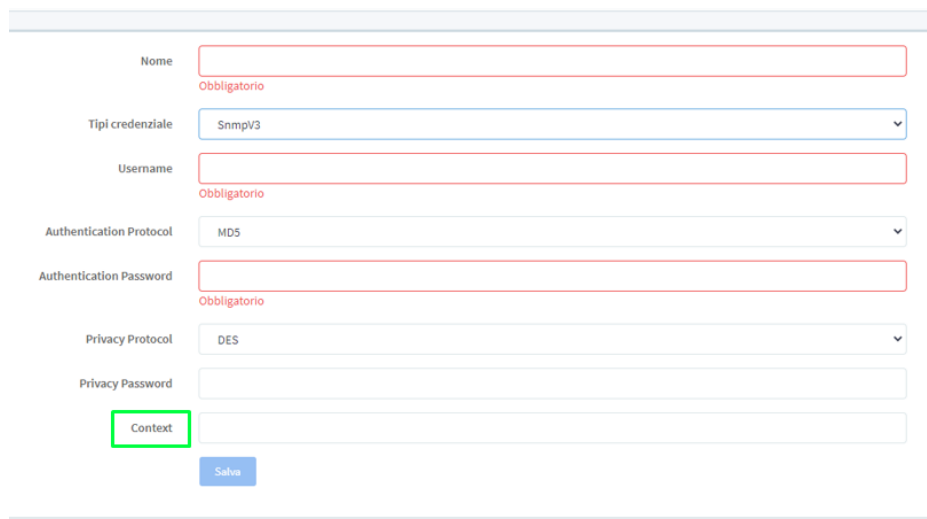
Read Password

Obbligatorio

SNMPV3

Enter the credential data and its authentication protocols (image 4). At the end of this section, we report some values for the **Context** field for the various printing brands (image 5).

Image 4



Nome

Obbligatorio

Tipi credenziale

Username

Obbligatorio

Authentication Protocol

Authentication Password

Obbligatorio

Privacy Protocol

Privacy Password

Context

Image 5

Vendor	Context name
Brother	user defined
Dell	
Epson	EPSON
Fuji Xerox	
HP	Jetdirect
Konica Minolta	user defined
Kyocera	
Lanier	GWNCs
Lexmark	
Ricoh	GWNCs
Samsung	
Sharp	mfpdirect
Toshiba	MFP
Xerox	

Integration with Papercut

One of the most important new features of the new DCA 4 is the integration with Papercut data.

Papercut is a print management software used by customers around the world to minimize waste and enjoy a secure and intuitive printing experience. Present in over **195 countries**, it is a **print management and monitoring** solution suitable for all types of customers, from large to small businesses.

In order to use Papercut data, you first need to know how to configure the export of Papercut data at the following address:

<https://www.papercut.com/support/resources/manuals/ng-mf/applicationserver/topics/configure-data-integrations.html>

It is also important that the **MpsMonitor.Dca.Client** service user (by default, "Network Service") can access the folder where the data is exported. The destination folder must be created before you configure Papercut export.

To add permissions to the folder, follow this path:

- go to *folder properties > security > edit*
- click on *advanced > search now* and select the **Network Service** user (image 1)
- check the box that gives editing permissions to the user (image 2)

Image 1

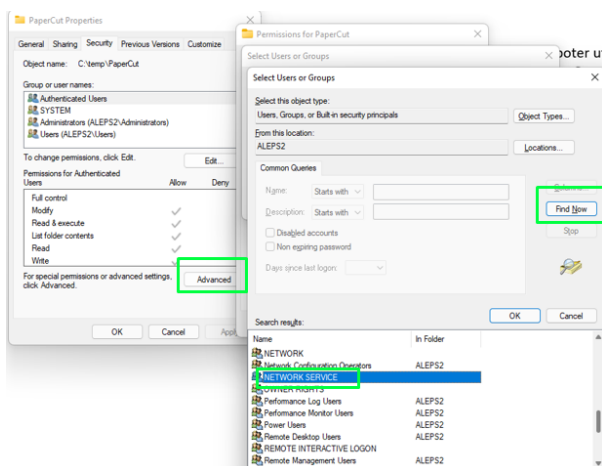
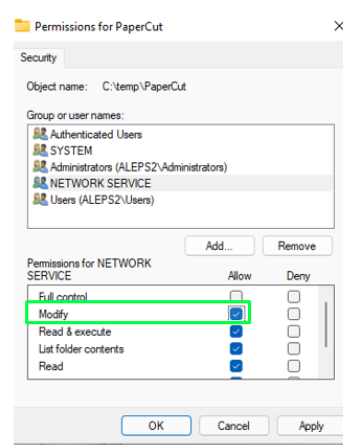


Image 2



At this point, to activate the integration connect to the MPS Monitor portal, select the customer for whom you want to activate it and enter the Customer **data page** (image 3). In the **Other Services** pane, press the **Activate** button on the papercut service (image 4).

Image 3

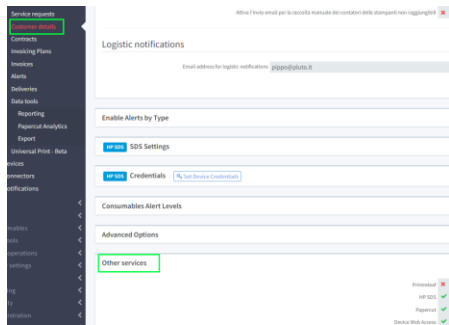
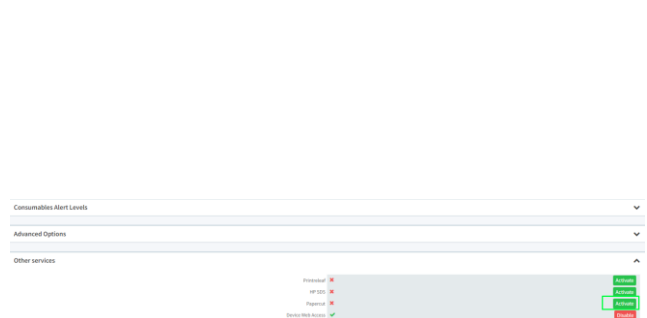


Image 4



Enter the path to the folder where the .csv files containing the usage data of the Papercut installation are exported and press **Enable**. You can also specify the field separator and the CSV string qualifier (image 5).

Image 5

Files folder

Enter the path to the folder where the .csv files containing the usage data of the PaperCut installation are exported (ex: C:\temp\exportPaperCut\)

CSV separator

Csv string qualifier

Once activated, a new entry will be available in the menu **Customer > Data Tools > Papercut Analytics** (image 6). In this section, users can enjoy an immersive **Business Intelligence** experience on their **Papercut** data, brought by **MPS Monitor** to the **Cloud** without licenses and additional costs.

The Papercut Analytics section contains two parts:

- **System reports:** Papercut reports published by MPS Monitor. These are ready-to-use reports that can only be opened and used in read-only mode.
- **Your reports:** this is your private area. You can create, edit and delete reports. Each user in MPS Monitor has their own personal area (image 7).

Image 6

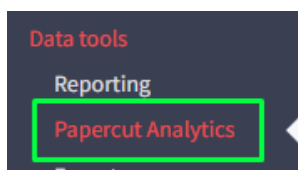
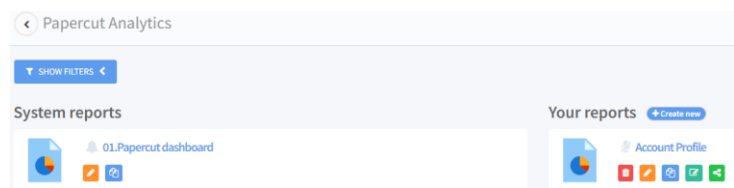
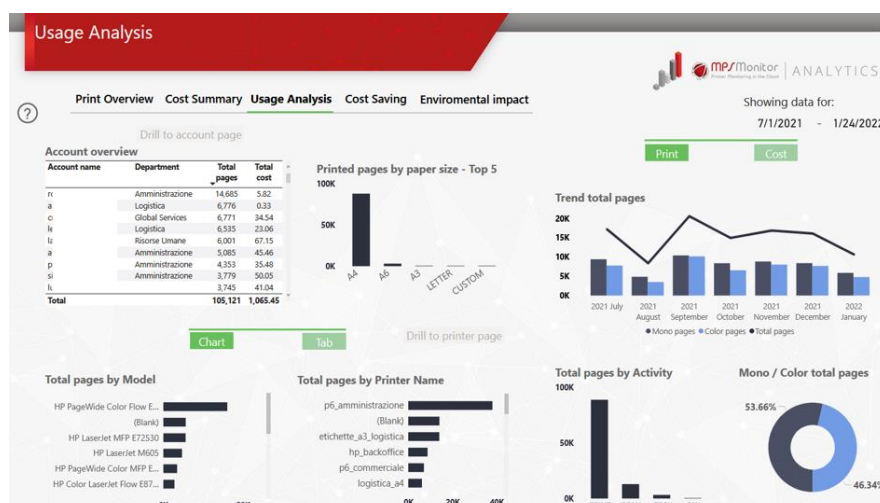


Image 7



So, in the predefined reports section, dealers and customers have access to dashboards and reports that take advantage of the advanced features of **Microsoft Power BI**®, an award-winning industry-leading business intelligence platform. Drawing **insights** for:

- Manage **print patterns** and identify **hidden costs**
- Identify **areas for improvement**
- Track **page volumes** by user, device, and department
- Reduce **environmental impact**
- Analyze **trends and** identify inefficiencies



Device Web Access

Another important novelty of the new DCA 4 is the ability to allow **access to the device web page** directly from the MPS Monitor portal without the need for a remote connection to the customer's network. Access is fast, secure, and transparent and works with all devices managed on the portal without distinction between brands and models.

This feature is only enabled for:

- Customers where DCA 4 is installed
- Devices monitored by DCA 4 (visible through Affinity)
- Users who have **two-factor authentication enabled** (or AD integration)



It is available on all devices owned directly by a dealer or customer - **NOT** on those belonging to the subdealer. Only top-level support operators can access DWA on all subdealers.

1. 2-factor authentication

The two-factor authentication that as seen above is fundamental for enabling the Web Access Device, is an authentication method that is based on the use of different authentication methods. Activating this feature will require an additional "disposable" code (OTP code). The code is generated through an application associated with the user through a "shared key".

How to enable 2-factor authentication?

- Go to the personal area > Edit profile (image 1)
- On the page that opens click on the button "Enable 2-factor authentication" in the upper right (image 2)
- At this point follow the procedure indicated in the menu that opens (image 3) and enter the generated OTP code
- Once created, disconnect and re-enter to make the procedure active

Image 1

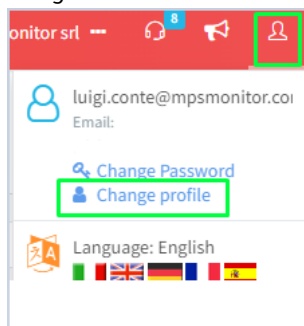


Image 2

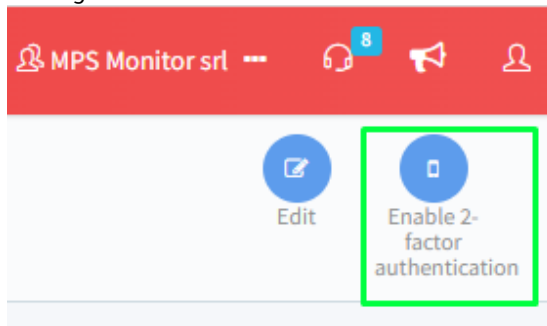


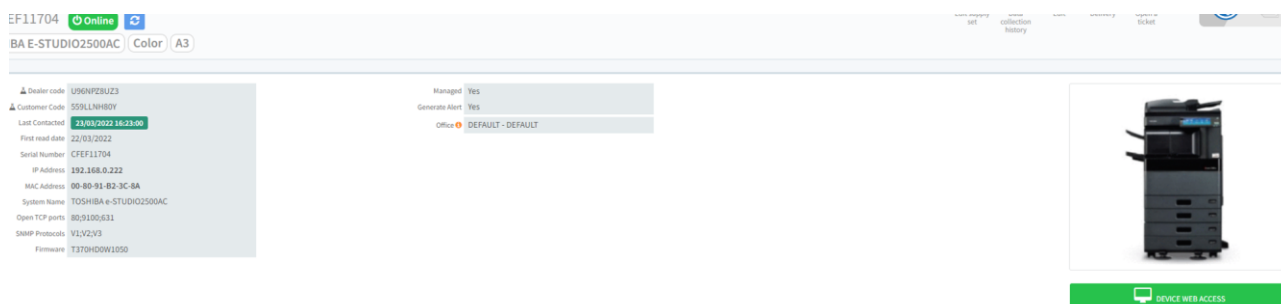
Image 3



2. Device Web Access requirements control

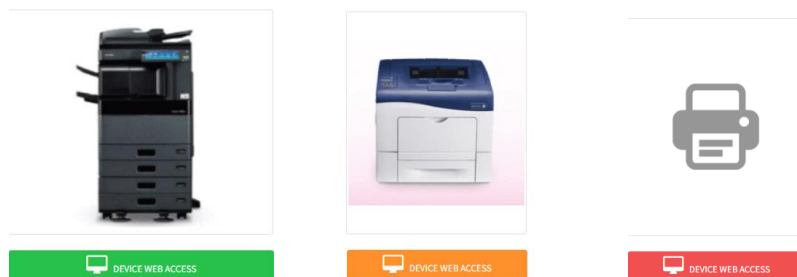
The first step is to click on the **Device Web Access** button on the right under the device image (image 4).

Image 4



At this point it's worth mentioning that the aforementioned button may appear in **different colors** depending on the outcome of a preliminary check of the system that verifies that all the requirements are met. Specifically, the button will appear in three different colors:

- **green**, if all the necessary requirements for connection to the printer are met
- **orange**, if some of the required requirements are not fully met. Remote connection is possible but may take longer than expected or fail
- **red** if one or more of the required requirements are not met. The Remote connection is not possible in this case



Clicking on the **Device Web Access** button will open a new page. Within this page by clicking **see all requirements** ([See all the requirements](#)), you will be faced with a list of requirements with OK status (**OK**) if the requirements are met, Warning status (**WARNING**) if they are not fully met, KO status (**KO**) if they are not met. Click on the info keypad (**i**) next to the requirement, to get more information about the requirement in question. In the next two sub-chapters we will see in detail 2 basic requirements: **Device - Date of Last communication** and **Connector - MQTT Protocol**.

Device Web Access

Tramite questa funzione e' possibile effettuare un collegamento remoto all'Embedded Web Server del dispositivo

Tutti i requisiti richiesti sono soddisfatti. E' possibile effettuare il collegamento remoto.

Il collegamento remoto verrà effettuato tramite il seguente connettore
 PC-051 (03beeca-2750-4133-4f0-24897479770a)
 all'indirizzo
<http://192.168.0.222:80> [Visualizza](#)

Requisito	Stato
Account - Numero di collegamenti remoti attivi	OK
Dispositivo - Data ultima comunicazione	OK
Cliente - Stato servizio collegamento remoto	OK
Connettore - Affidabilità	OK
Connettore - Stato servizio collegamento remoto	OK
Connettore - Data ultima comunicazione	OK
Connettore - Protocollo MQTT	OK

[Visualizza requisiti](#)

[Apri collegamento remoto](#) Chiudi

Device Web Access

Through this function it is possible to make a remote connection to the Embedded Web Server of the device

Some of the required requirements are not fully met. Remote connection is possible but may take longer or fail.

The remote connection will be made via the following connector
 ALP5C2 (a966d618b-20a3-4515-9f71-e0908539905)
 at the web address
<http://192.168.230.247:80> [View](#)

Requirement	State
Account - Number of active remote connections	OK
Device - Date of last communication	WARNING
Customer - Customer - Remote connection service status	OK
Connector - Affinity	OK
Connector - Remote Connection Service Status	OK
Connector - Date of last communication	WARNING
Connector - MQTT Protocol	OK

[View requirements](#)

Reason

Insert the reason for opening the remote session (like the ticket number or the activity performed). This is required for Audit reasons and will be visible to the customer.

[Open remote connection](#) Close

Device Web Access

Through this function it is possible to make a remote connection to the Embedded Web Server of the device

One or more required requirements are not met. Remote connection is not possible.

The remote connection will be made via the following connector
 ALP5C2 (a966d618b-20a3-4515-9f71-e0908539905)
 at the web address
<http://192.168.230.247:80> [View](#)

Requirement	State
Account - 2 Factor Authentication	KO
Account - Number of active remote connections	OK
Device - Date of last communication	OK
Customer - Customer - Remote connection service status	OK
Connector - Affinity	OK
Connector - Remote Connection Service Status	OK
Connector - Date of last communication	OK
Connector - MQTT Protocol	OK

[See requirements KO](#)

[Close](#)

2.1 Device Requirement - Date of last communication

If this requirement is in Warning state (**WARNING**) it means that the device has not **communicated within the last 10 minutes**. In this case, the device will be classified as **Not online** (image 11). So, click on the **Scan now** button (), placed next to the device status, to make an immediate scan of the device data and, subsequently, disconnect and reconnect to the Portal to allow a correct synchronization. If the process is successful, the device will appear as **Online** (image 12) and, by clicking the **Device Web Access** button again, the status of the requirement will appear in OK state ().

Image 11

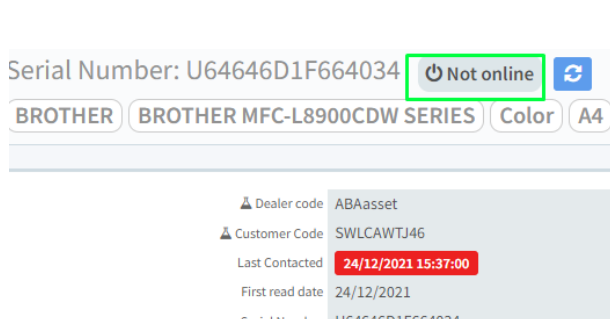
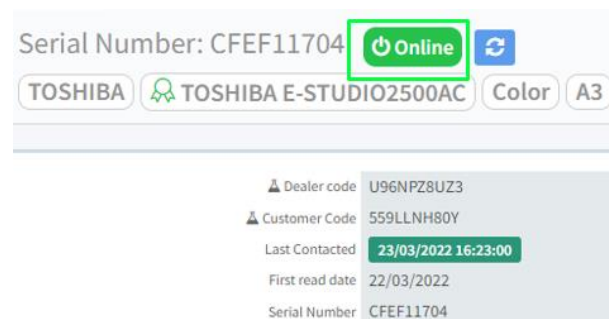


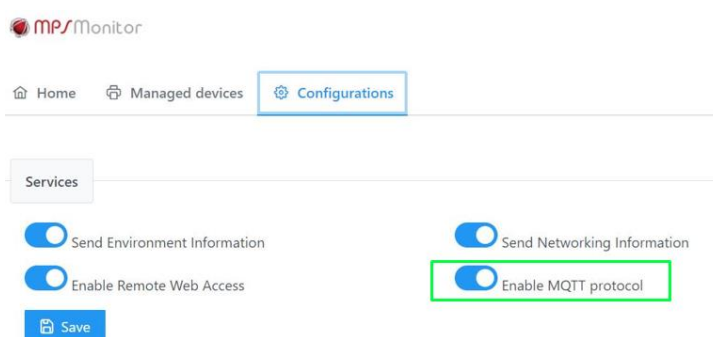
Image 12



2.2 Connector Requirement - MQTT Protocol

If the MQTT protocol requirement is not OK (), you will need to access **the connector configuration** page through link <http://localhost:12312/home> (chapter *Performing DCA 4 installation*) - **Configurations** tab - to verify that the protocol is enabled and, If not, enable it.

Image 13



3. Door control

On the page that opens by clicking on the Device Web Access button you can change the ports. Clicking on **Edit** next to the address (image 14) opens a submenu where you can choose another port from the options in the drop-down menu (image 15).

Image 14

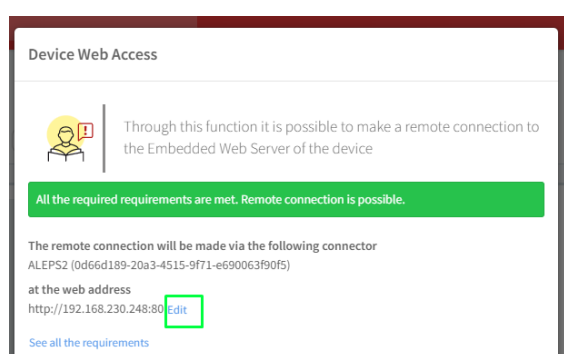
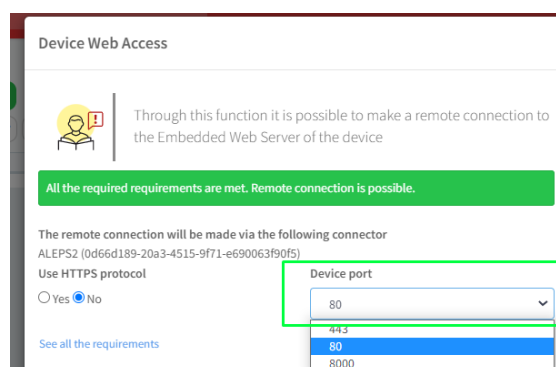


Image 15



4. Link to the device web page

Once the requisites and ports have been checked, at this point by clicking on the **open remote connection button** (image 16), after a wait of about 30 sec - 1 min, you will access the device's web page (image 17).

Image 16

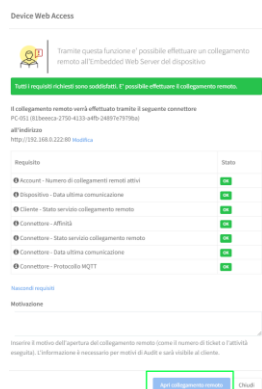
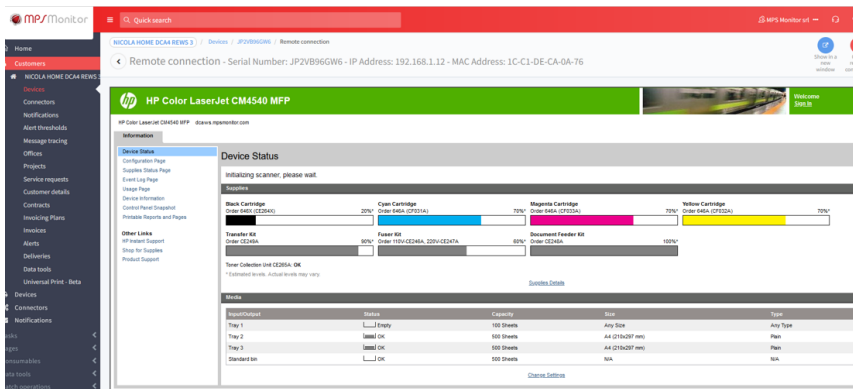


Image 17

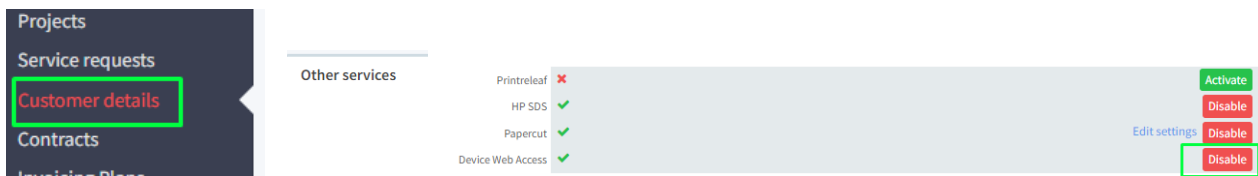


5. How to disable Device Web Access

Device Web Access is enabled on all customers by default. However, it can be deactivated both at the customer level (via Portal) and at the DCA level (locally).

To turn it off at customer level, follow the path **Customer Data > Other Services > Disable Device Web Access** as shown in image 18.

Picture 18



If you want to disable at the DCA level, access **the connector configuration** page through link <http://localhost:12312/home> (chapter Performing DCA 4 installation) and disable it from the Configurations submenu, as shown in image 19.

Picture 19

